

# Deep Dive: 3. Operators

## Concept Summary

Operators are special symbols that carry out arithmetic or logical computation.

## Coding Structure

```
result = value1 operator value2
```

## Example Code

```
x = 10
y = 3
print(x // y) # Floor division
print(x % y)  # Modulo
print(x > 5 and y < 5) # Logical AND
```

## Code Explanation

- Line 3: Floor division (//) divides 10 by 3 and chops off the decimal, resulting in 3.
- Line 4: Modulo (%) returns the remainder of 10 divided by 3, which is 1.
- Line 5: The 'and' operator checks if BOTH conditions are true. Since 10 > 5 and 3 < 5, it prints True.

## Practice Problems

1. Calculate and print the area of a rectangle with width 15 and height 7.
2. Write an expression that checks if a number is even by using the modulo operator (%).